CE4011 Structural Analysis GUI

Comprehensive User Manual

GUI-Based Static and Modal Postprocessing Extension of a 2D Frame-Truss Direct Stiffness Solver

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Course: CE4011 Structural Analysis Software Development
Python environment: CE4011, Python 3.11
Date: 19 June 2026

Scope of this manual: This document describes all user-facing features of the software: model input, generated examples, static analysis, modal analysis, result plots, result tables, diagnostics, saving/export, educational dynamic postprocessing utilities, and common troubleshooting. Installation details are kept in the separate Installation Manual.

1. Program overview

The CE4011 Structural Analysis GUI is an educational desktop program for linear elastic two-dimensional frame-truss structural analysis. It combines a Python Direct Stiffness Method backend with a Tkinter/Matplotlib graphical interface. The software is intended for learning, verification, and demonstration of structural-analysis software development, not for commercial structural design.

- Structural system: two-dimensional frame-truss models.
- Element types: frame elements with axial and flexural behavior; truss elements with axial-only behavior.
- Main analyses: linear static analysis and modal analysis.
- Postprocessing: deformed shapes, internal-force diagrams, station forces, mode shapes, story drift, roof displacement, result tables, CSV export, and solver diagnostics.
- Extended utilities: selected linear modal response-history and response-spectrum postprocessing tools.
- Verification support: generated examples, textbook-style modal examples, Ftool comparisons, Robot comparison data, automated tests, and GUI-backend verification scripts.

Important limitation: The software does not perform code design checks, full 3D analysis, shell/slab modeling, rigid diaphragm modeling, P-Delta analysis, material/geometric nonlinear analysis, or commercial load-combination design workflows.

2. Quick start

2.1 Starting the installed executable

For the packaged Windows distribution, keep the complete dist\CE4011_Solver folder together and run the executable inside that folder. Do not move the executable away from its adjacent _internal folder, because that folder contains the bundled Python runtime, Tcl/Tk, NumPy, SciPy, Matplotlib, and demonstration data.

```
dist\CE4011_Solver\CE4011_Solver.exe
```

2.2 Starting from source

For the source-code workflow, run the program from the repository root. Running from another directory can break relative paths for inputs, generated models, companion mass files, and result folders.

```
conda activate CE4011
python scripts\run_static_gui.py
```

2.3 First useful workflow

1. Launch the GUI.

2. Use File -> Open Generated Model and choose model_a_5story_unbraced.json.

3. Inspect the model view for nodes, elements, supports, loads, and modal masses.

4. Run Analyze -> Run Static Analysis.

5. Open Display -> Static Deformed Shape and Display -> Bending Moment Diagram.

6. Open Tables -> Nodal Displacements and Tables -> Support Reactions.

7. Run Analyze -> Run Modal Analysis and request four modes.

8. Open Display -> Mode Shapes and Tables -> Modal Frequencies / Modal Participation Factors.

3. Main window and interface layout

The main window is organized around a model display area, summary/status information, and menu-driven analysis and postprocessing actions. The GUI follows a typical structural-analysis workflow: define or load a model, inspect it, analyze it, display plots, open tables, and export selected results.

Area	Purpose
Menu bar	Access to file operations, model definition, analysis commands, display commands, result tables, selection tools, and help.
Model/result canvas	Displays model geometry, deformed shapes, force diagrams, mode shapes, drift plots, response plots, or other Matplotlib figures.
Summary/status panel	Reports loaded-model information, analysis status, result summaries, selected node/element data, and diagnostic messages where available.
Dialog windows	Used for defining materials, sections, nodes, supports,

Area	Purpose
	elements, loads, masses, units, and conventions.
Table windows	Show numerical results and usually allow CSV export for reporting or verification.

4. Menu and feature inventory

The following matrix summarizes the main user-facing features. Detailed explanations are provided in later sections.

Menu	Main actions	Typical use
File	New model; open JSON/XML/text model; open generated model; save model; save model as; save model package where available.	Create, load, reopen, or preserve model files.
Define	Materials; sections; nodes; elements; supports; nodal loads; member loads; thermal loads; support settlements; modal masses; axis/rigid offsets; units; diagram conventions.	Build or edit models using GUI forms.
Analyze	Run static analysis; run modal analysis; optional linear modal RHA/RSA utilities.	Solve the currently loaded model.
Display	Model view; static deformed shape; Hermite deformed shape; axial/shear/bending force diagrams; station-force plots; deformed slope profile; mode shapes; drift; roof/floor response; RHA/RSA plots.	Visualize results and engineering behavior.
Tables	Nodal displacements; support reactions; member-end forces; station forces; slopes; modal frequencies; participation; response factors; roof displacement; story drift; solver diagnostics.	Inspect numerical results and export CSV files.
Select	Select nodes/elements; select all; clear selection.	Inspect or highlight parts of larger models.
Help	Workflow notes, usage summaries, and model guidance where implemented.	Recall commands and interpretation notes inside the GUI.

5. Input routes and data organization

The program supports five model input routes. After loading, all routes are converted into a common backend model dictionary before a Structure object is created. This means a JSON model, XML model, text-deck model, GUI-defined model, and generated model all use the same solver path after validation.

Input route	Description	When to use
JSON	Structured model files with geometry, materials, sections, elements, boundary conditions, loads, and optional modal data.	Recommended default for reproducible examples and saved GUI models.
XML	Structured XML representation of the model.	Useful when testing alternative structured file formats.
Text-deck	Compact keyword-style input route	Useful for small examples and

Input route	Description	When to use
	retained for reproducible tests and verification examples.	regression tests.
GUI forms	Interactive creation/editing of materials, sections, nodes, supports, elements, loads, and masses.	Useful for building simple models directly in the interface.
Generated models	Parametric frame examples such as five-story unbraced, ten-story unbraced, and ten-story braced frames.	Useful for demonstration, modal examples, and trend verification.

Canonical model path: Once a model is converted to the backend representation, the same static, modal, plotting, table, and export logic is used regardless of how the model was originally created.

6. Units and numerical conventions

The default unit convention is force in N, length in m, mass in kg, time in s, temperature in C, and rotation in rad. Unit settings are used as metadata and labels in the GUI, tables, and plots. They do not automatically rescale existing numerical input values.

Quantity	Default unit	Notes
Force	N	Use kN only if the numerical values are consistently entered in kN and labels are changed accordingly.
Length	m	Geometry, displacements, member lengths, and offsets must be consistent.
Mass	kg	Modal masses should be physical lumped masses, not weights.
Time	s	Used for periods and response-history utilities.
Temperature	C	Used for thermal-loading metadata/input.
Rotation	rad	Rotational DOFs and slopes are reported in radians.

No automatic conversion: Changing displayed units does not convert previously entered numbers. Users must maintain a consistent unit system.

7. Model definition through the GUI

7.1 Recommended manual modeling order

1. Create a new model or clear the current model.

2. Define materials.

3. Define sections.

4. Define nodes and coordinates.

5. Define supports or prescribed restraints.

6. Define frame and/or truss elements.

7. Define nodal loads, member loads, thermal loads, or support settlements as needed.

8. Define modal masses if modal analysis will be performed.

9. Inspect the model view before solving.

10. Run static analysis first; run modal analysis only after modal mass data are available.

7.2 Materials

Materials store mechanical properties used by frame and truss elements. A typical material record includes an identifier/name and elastic properties such as Youngs modulus. Preset materials may be available for quick model creation. Confirm that material units are consistent with the geometry and load units.

7.3 Sections

Sections store geometric properties used by elements. Frame elements normally require area and bending inertia. Truss elements require axial area. Preset sections may be used for quick examples. Section values should be physically consistent with the selected material and unit system.

7.4 Nodes

Nodes define the geometry of the 2D model. Each node normally has x and y coordinates and three structural degrees of freedom: horizontal translation, vertical translation, and in-plane rotation. Use clear node IDs and avoid duplicate coordinates unless intentional.

7.5 Supports and restraints

Supports define restrained degrees of freedom. Common support conditions include fixed, pinned, roller, and free DOFs, depending on the implemented dialog options. The program applies boundary conditions during static analysis and classifies free/restrained DOFs for diagnostics.

7.6 Frame elements

Frame elements connect two nodes and include axial and flexural stiffness. They are suitable for beams and columns. Frame elements can participate in member loads, thermal loads, support settlements, force recovery, bending moment diagrams, shear force diagrams, and deformed-shape plotting.

7.7 Truss elements

Truss elements connect two nodes and carry axial force only. They are suitable for diagonal braces or axial members. Unsupported member-load cases on truss elements are rejected because transverse distributed or point loads require flexural behavior.

7.8 Axis offsets and rigid-end style offsets

The program includes limited two-dimensional axis-offset or rigid-end-offset style metadata where supported. Offsets are intended for local-y frame offset demonstrations and plotting/analysis workflows implemented in the backend. Unsupported offset combinations, especially with truss members, are rejected to avoid misleading results.

8. Loads and imposed effects

Input type	Description	Typical output affected
Nodal loads	Forces or moments applied directly to node DOFs.	Displacements, reactions, member forces, diagrams.
Frame point loads	Supported point loads on frame elements, typically in local member directions where implemented.	Equivalent nodal loads, member-end forces, station forces, diagrams.
Full-span UDLs	Uniformly distributed loads along supported frame elements.	Equivalent nodal loads, shear and moment diagrams, station forces.
Thermal loads	Thermal expansion effects inherited from the earlier assignment workflow.	Static displacement/reaction/member force outputs.
Support settlements	Prescribed support displacement effects inherited from the earlier assignment workflow.	Static displacement/reaction/member force outputs.
Modal masses	Lumped masses used for modal analysis.	Frequencies, mode shapes, participation factors, effective modal masses.

8.1 Nodal loads

Nodal loads are applied directly to a selected node and DOF. Use the correct sign convention for the global coordinate system. For a downward vertical load in an N-m system, the vertical force is usually negative.

8.2 Member loads

Supported member loads include frame-element point loads and full-span uniformly distributed loads. Member loads are converted into equivalent nodal loads and also used for local force recovery so that internal-force diagrams and station forces reflect the load distribution along the member.

Unsupported member loads: Partial UDLs and unsupported truss/member-load combinations are not part of the final supported core unless explicitly implemented in the code. Rejected cases should be treated as intentional input validation, not solver failure.

8.3 Thermal loads

Thermal loads are static imposed effects. They are included in static analysis but are not modal masses. They should not be confused with dynamic mass or response-history input.

8.4 Support settlements

Support settlements prescribe displacement effects at restrained or support-related DOFs depending on the model. They are used in static analysis and can generate reactions and member forces even without ordinary nodal loads.

9. Opening generated frame models

Generated frame models provide complete example structures for analysis and verification. They are useful for confirming that static and modal workflows operate on larger structural systems, not only single-element examples.

Generated example	Purpose	Important notes
model_a_5story_unbraced.json	Main static/modal demonstration model.	Use with its companion modal mass file. Good for Ftool and Robot

Generated example	Purpose	Important notes
		comparison.
10-story unbraced frame	Larger unbraced frame example.	Useful for drift, roof displacement, and modal trend checks.
10-story braced frame	Larger braced frame with diagonal truss braces.	Useful for comparing stiffness, drift reduction, and frequency increase relative to the unbraced model.

9.1 Companion modal mass files

Some generated models store modal mass data in a separate companion file. Keep the companion mass file in the expected location with the main model file. For example:

```
model_a_5story_unbraced.json
model_a_5story_unbraced_masses.json
```

If the companion file is missing, modal analysis may fail or use documented fallback behavior depending on the workflow. For physically meaningful modal results, always verify that the mass source is intentional.

10. Static analysis

10.1 Running static analysis

1. Load or define a stable model.

2. Inspect geometry, supports, and loads in Model View.

3. Use Analyze -> Run Static Analysis.

4. Confirm the analysis status in the summary panel.

5. Open static plots and tables from Display and Tables menus.

10.2 What the static solver does

The static solver assembles the global stiffness matrix from element stiffness contributions, applies boundary conditions and prescribed displacement effects, solves the reduced active-DOF system, recovers the full displacement vector, computes support reactions, and recovers local member-end forces. These results feed deformed-shape plots, force diagrams, tables, drift calculations, and diagnostics.

10.3 Common static outputs

- Nodal displacement table.
- Support reaction table.
- Member-end force table.
- Element station-force table.
- Static deformed shape.
- Hermite deformed shape where available.
- Axial force diagram.
- Shear force diagram.
- Bending moment diagram.
- Story drift and roof displacement for multi-story frames.

- Solver diagnostics.

Deformed shape scale: Plots use visual magnification. A large-looking deformed shape does not imply a large physical displacement. Read actual values from the displacement table or summary panel.

11. Internal-force diagrams and station forces

11.1 Axial force diagram

The axial force diagram displays member axial force along the member geometry. It is most important for truss braces, columns, and axial response in frame members.

11.2 Shear force diagram

The shear force diagram displays transverse shear along frame elements. For point loads and UDLs, the station-force output should be used to inspect intermediate values along the span.

11.3 Bending moment diagram

The bending moment diagram displays flexural response along frame elements. The plotted side/sign is a display convention and may differ from Ftool or another program, but the numerical values remain defined by the solver convention.

11.4 Section force stations

Station forces report axial force, shear force, and bending moment at multiple locations along an element. Use Tables -> Element Station Forces for numerical station data. This is especially useful for UDL and point-load cases where values vary along the member, not only at the ends.

11.5 Diagram conventions

The GUI may provide diagram display convention settings to control plotted side or sign for visual comparison. Changing a display convention should not change the underlying numerical member forces.

12. Modal analysis

12.1 Required mass data

Modal analysis requires modal mass data. Modal masses can be provided explicitly in the model, through companion mass files, by generated-model definitions, or through GUI-defined modal mass records. Without meaningful mass data, modal frequencies and participation quantities are not physically meaningful.

12.2 Running modal analysis

1. Open a model with modal masses.

2. Run static analysis first if the workflow or display sequence requires static results.

3. Use Analyze -> Run Modal Analysis.

4. Select the number of modes to compute/display.

5. Open mode-shape plots and modal tables.

12.3 Massless DOF handling

In 2D frame models, rotations and some vertical DOFs may have zero assigned mass. A full mass matrix with massless DOFs can be singular. The modal solver separates free DOFs into massive and massless sets, statically condenses the massless DOFs, solves the reduced eigenvalue problem, and recovers full free-DOF mode shapes for display and postprocessing.

12.4 Modal outputs

- Circular frequency, omega, in rad/s.
- Natural frequency, f, in Hz.
- Period, T, in seconds.
- Mode shapes.
- Modal participation factors.
- Effective modal masses.
- Modal mass ratios and cumulative mass ratios.
- Modal base shear and modal moment style response quantities where implemented.
- DOF classification information for massive and massless DOFs.

Mode-shape sign: Eigenvector sign is arbitrary. A mode shape may appear inverted compared with Robot, Ftool, lecture notes, or another program. Compare frequencies, periods, participating masses, and qualitative mode-shape patterns, not raw sign alone.

13. Linear modal RHA and RSA utilities

The program includes selected educational dynamic postprocessing utilities based on computed modal properties. These are controlled extensions for learning and demonstration. They are not nonlinear seismic analysis, not design-code workflows, and not substitutes for commercial dynamic design software.

13.1 Ground-motion and response-history workflow

1. Open a model with valid modal mass data.

2. Run modal analysis first.

3. Select the response-history utility from the Analyze or Display workflow where implemented.

4. Choose the ground-motion record file when prompted.

5. Set damping ratio, units, and number of modes as required by the dialog.

6. Inspect roof displacement, floor displacement, story drift, modal coordinate, and selected-node response plots.

13.2 Response-spectrum workflow

1. Run modal analysis first.

2. Open the response-spectrum utility.

3. Generate or load the spectrum as implemented.

4. Choose combination method outputs such as ABSSUM, SRSS, and CQC.

5. Inspect peak story drift, roof response, modal force, and combined response plots/tables where available.

13.3 Typical dynamic postprocessing outputs

- Ground-motion acceleration plot.
- Roof displacement history.
- Floor displacement histories.
- Story drift histories and peak story drift.
- Modal coordinate histories.
- Selected node response history.
- Elastic response-spectrum plot.
- RSA peak story response.
- ABSSUM, SRSS, and CQC combination comparison.
- Modal force or modal response factor tables where implemented.

14. Display menu - plot reference

Display item	Requires	What it shows
Model View	Loaded/defined model.	Nodes, elements, supports, loads, labels, modal mass markers, settlements, and offset indicators where implemented.
Static Deformed Shape	Static analysis.	Magnified deformed geometry based on solved displacements.
Hermite Deformed Shape	Static analysis on frame elements.	Smooth beam-column deformation interpolation where implemented.
Axial Force Diagram	Static analysis.	Axial member force diagram on model geometry.
Shear Force Diagram	Static analysis.	Frame-element shear force diagram.
Bending Moment Diagram	Static analysis.	Frame-element bending moment diagram.
Section Force Stations	Static analysis and station recovery.	N/V/M values at stations along members.
Deformed Slope Profile	Static analysis and slope recovery.	Slope or rotation-style deformation profile where implemented.
Mode Shapes	Modal analysis.	Selected natural mode shape, often with scale control.
Modal Frequency Plot	Modal analysis.	Frequency or period summary by mode.
Story Drift Profile	Static or dynamic result depending on workflow.	Interstory drift distribution for frame models.
Roof/Floor Response	Static, modal, RHA, or RSA depending on plot.	Roof displacement or floor response quantities.
RHA/RSA Plots	Modal analysis plus dynamic utility run.	Response-history or response-spectrum postprocessing figures.

15. Tables menu - numerical result reference

Table	Requires	Use
Nodal Displacements	Static analysis.	Read actual displacement and rotation values by node/DOF.
Support Reactions	Static analysis.	Check support reactions and global

Table	Requires	Use
		equilibrium.
Member-End Forces	Static analysis.	Inspect local/global element end forces.
Element Station Forces	Static analysis and station recovery.	Inspect N/V/M along elements, especially under member loads.
Element Deformed Slopes	Static analysis and slope recovery.	Inspect slope/rotation interpolation data where available.
Modal Frequencies	Modal analysis.	Read omega, frequency, period, and mode count.
Modal Participation Factors	Modal analysis.	Read participation factors, effective masses, mass ratios, and cumulative mass.
Modal Response Factors	Modal/RSA utilities.	Inspect modal response quantities where implemented.
Roof Displacement	Static or dynamic postprocessing.	Read roof response for multi-story frame examples.
Story Drift	Static or dynamic postprocessing.	Read interstory drift values.
Solver Diagnostics	Loaded model or completed analysis.	Inspect nodes, elements, DOFs, stiffness size, nonzeros, density, bandwidth, and RCM-style estimates where implemented.

15.1 CSV export

Most result tables can be exported as CSV files. Use descriptive names that identify the model, analysis type, and table. For example, braced_modal_frequencies.csv is clearer than output1.csv. CSV export is useful for verification, plotting outside the GUI, and including selected values in reports.

16. Selection tools

Selection tools help inspect large generated frames. Depending on the active view, users can select nodes, select elements, select all nodes/elements, or clear the selection. Selection is mainly a visualization and inspection tool; it should not be treated as changing the structural model unless a specific editing dialog says so.

Action	Typical purpose
Select node	Inspect coordinates, support condition, loads, modal mass, or response data for a node.
Select element	Inspect element connectivity, type, material, section, offsets, loads, and force results.
Select all nodes/elements	Highlight all objects for visual inspection or confirming selection behavior.
Clear selection	Return to an uncluttered model/result view.

17. Saving, reopening, and model packages

17.1 Saving a model

Use the File menu to save the current model. JSON is the recommended default for reproducible saved models. After saving, reopen the model once and check that geometry, supports, loads, sections, materials, and modal masses were preserved correctly.

17.2 Saving companion modal mass data

If modal masses are stored in a companion file, save or preserve that file together with the main model file. Do not move one without the other. A model can look correct geometrically but still fail modal analysis if its companion mass file is missing.

17.3 Generated outputs

Scripts and some GUI/export workflows write generated outputs under results/ or selected result folders. Generated outputs may include PNG figures, CSV files, diagnostics, or comparison data. These outputs are reproducible artifacts and may be regenerated from the model and scripts when needed.

18. Solver diagnostics

The solver diagnostics table reports computational information that connects the model to the numerical backend. Diagnostics are useful for debugging, teaching, and confirming that the DOF system size is reasonable.

Diagnostic item	Meaning
Number of nodes/elements	Basic model size and connectivity summary.
Total/free/restrained DOFs	Degree-of-freedom count after supports and restraints are applied.
Active stiffness matrix size	Size of the reduced system solved for active DOFs.
Nonzero stiffness entries	Sparse matrix storage and assembly indicator.
Matrix density	Fraction of nonzero entries compared with a dense matrix.
Semi-bandwidth/full bandwidth	Bandwidth-related property relevant to matrix storage/solution efficiency.
RCM-estimated bandwidth	Optional reordered bandwidth estimate where implemented.
Massive/massless DOF classification	Modal-analysis classification of DOFs with or without assigned mass.

19. Verification examples included or supported

The project includes several verification examples and comparison workflows. These are useful for users who want to check whether the installation and solver behavior are consistent with known reference results.

Verification case	Reference/idea	What it checks
Cantilever beam static verification	Hand/Ftool-style beam solution.	Displacements, reactions, shear force, bending moment.
Simply supported beam with UDL	Hand/Ftool-style beam solution.	Support reactions, shear diagram, moment diagram, load recovery.
Thermal load and support settlement examples	Earlier assignment-style checks.	Imposed effects and static result recovery.
CE586 modal benchmark examples	Lecture/textbook-style modal examples.	Frequencies, periods, participation, effective modal mass, modal forces.
Five-story generated frame static check	Ftool/commercial-style comparison.	Generated model stiffness, lateral loads, displacement, internal forces.
Five-story Robot modal comparison	Autodesk Robot modal analysis.	Generated-model modal frequencies and mode-shape trends.
Ten-story braced/unbraced comparison	Engineering trend validation.	Braces reduce drift/displacement and increase frequency.
Automated pytest suite	Regression/unit tests.	Static, modal, input, plotting, postprocessing, saving, import stability.
GUI-backend verification manifest	Scripted GUI/backend consistency cases.	Confirms GUI routes call the same backend behavior.

20. Complete example workflow A - cantilever beam

1. Launch the GUI.

2. Open the cantilever beam example from the examples/input folder or generated/example menu if available.

3. Confirm that one end is fixed and the other end carries the applied load.

4. Run Analyze -> Run Static Analysis.

5. Open Display -> Static Deformed Shape.

6. Open Display -> Shear Force Diagram and Display -> Bending Moment Diagram.

7. Open Tables -> Nodal Displacements and Tables -> Support Reactions.

8. Compare the reported shear and moment magnitudes with the verification reference.

9. Use File -> Save Model As if you want to preserve a modified copy.

Interpretation: A cantilever is a compact check for boundary conditions, load sign, reaction recovery, shear force, bending moment, and deformed-shape plotting.

21. Complete example workflow B - five-story modal frame

1. Launch the GUI.

2. Open model_a_5story_unbraced.json using File -> Open Generated Model or File -> Open JSON model.

3. Keep model_a_5story_unbraced_masses.json in the expected companion location.

4. Inspect Model View for fixed bases, beams, columns, lateral loads, and mass indicators.

5. Run Analyze -> Run Static Analysis.

6. Display Static Deformed Shape and Bending Moment Diagram.

7. Open Tables -> Nodal Displacements and Support Reactions to check static values.

8. Run Analyze -> Run Modal Analysis and request four modes.

9. Display Mode 1 through Mode 4.

10. Open Modal Frequencies and Modal Participation Factors tables.

11. Export modal tables as CSV if needed.

12. Open Story Drift, Roof Displacement, or Solver Diagnostics if relevant.

Interpretation: For the five-story frame, the first mode should behave like a lateral sway mode. Frequency and mass participation are more important than the arbitrary sign of the plotted eigenvector.

22. Complete example workflow C - braced versus unbraced frames

1. Open the ten-story unbraced generated frame.

2. Run static analysis and record roof displacement and maximum story drift.

3. Run modal analysis and record the first natural frequency.

4. Open the ten-story braced generated frame.

5. Repeat static and modal analyses.

6. Compare the two cases.

The expected engineering trend is that braces increase lateral stiffness, reduce roof displacement, reduce story drift, and increase the first natural frequency. Exact values depend on the current model data, but the direction of change should be structurally reasonable.

23. Error handling and common warnings

Symptom/message	Likely cause	Recommended action
GUI does not launch	Wrong environment, missing Tkinter, or wrong working directory.	Use the Installation Manual. Run from repository root or keep the full dist folder intact.
No module named gui	src/ package not on Python path or script run from wrong location.	Run python scripts\run_static_gui.py from repository root. Do not run from inside scripts/ or src/.
Failed to load Python DLL	Executable moved away from _internal or app run from build/ instead of dist/.	Run the complete dist\CE4011_Solver folder. Do not copy only the exe.
Model opens but modal analysis fails	Missing or invalid modal mass data.	Check explicit modal masses or restore the companion mass file.
Static model is singular/unstable	Insufficient restraints, mechanism, bad connectivity, or invalid element definition.	Inspect supports, connectivity, duplicate nodes, and element properties.
Text-deck load fails	Syntax error, missing field, invalid keyword, or invalid reference.	Use the reported line/keyword if available and compare with working examples.
Duplicate ID or invalid reference	Node/element/material/section references do not match.	Check IDs and ensure referenced objects exist before use.
Unsupported truss/member load	A load type requiring flexural behavior was assigned to a truss or unsupported case.	Use a frame element or a supported load type.
Deformed shape looks too large	Plot magnification, not physical displacement.	Read values from tables and adjust display scale if available.
Force diagram sign differs from Ftool	Different plotting/sign convention.	Compare magnitudes and numerical table values; adjust display convention if available.
Units appear wrong	Unit labels changed without converting numerical inputs.	Use a consistent input unit system; unit metadata does not rescale values.
Old results appear after failed analysis	Stale results should be cleared by the app, but confusion can occur after	Rerun analysis after fixing model; reload model if uncertain.

Symptom/message	Likely cause	Recommended action
	errors.	

24. Best practices for reliable use

- Run the software from the repository root in source mode.
- Keep companion modal mass files beside their model files.
- Use simple verification cases before interpreting larger models.
- Check model view before analysis; most errors are input/connectivity problems.
- Run static analysis before interpreting static plots or tables.
- Run modal analysis only when mass data are physically meaningful.
- Use tables for numerical values; use plots for interpretation and communication.
- Export CSV files with descriptive names.
- Do not interpret educational RHA/RSA utilities as nonlinear seismic design checks.
- Save and reopen important models to confirm persistence.

25. Limitations

The software is intentionally limited in scope. It is an educational linear elastic 2D frame-truss analysis program. The following features are outside the current supported scope:

- Full 3D frame analysis.
- Shell, plate, slab, or continuum finite elements.
- Rigid diaphragm constraints.
- Material nonlinearity.
- Geometric nonlinearity and P-Delta effects.
- Staged construction.
- Commercial load-combination workflows.
- Design-code checks.
- Complete nonlinear response-history analysis.
- Mesh generation and contour visualization.
- Automatic unit conversion of existing model values.

26. Future extension ideas

- 3D frame elements and 6-DOF-per-node formulation.
- Shell or slab elements.
- Rigid diaphragm constraints for building models.
- Partial-span member-load formulas.
- P-Delta analysis and geometric stiffness.
- Nonlinear material models and nonlinear response-history analysis.
- Design-code load combinations and member checks.
- More robust CAD-style model editing.
- Reusable shape-function libraries, Gauss quadrature, isoparametric elements, meshing, and contour plotting.
- Improved packaging and cross-platform distribution.

Appendix A. Glossary

Term	Meaning
DSM	Direct Stiffness Method. Matrix method used to assemble and solve structural equilibrium equations.
DOF	Degree of freedom. Translational or rotational displacement component at a node.
Frame element	Two-node member with axial and flexural stiffness.
Truss element	Two-node axial-only member.
Modal mass	Lumped mass used for dynamic/modal analysis.
Massless DOF	Free degree of freedom with zero assigned modal mass.
Static condensation	Reduction procedure eliminating massless DOFs before solving the modal eigenproblem.
Mode shape	Eigenvector describing a natural vibration pattern. Sign and normalization are arbitrary.
Participation factor	Quantity measuring how strongly a mode participates in a selected excitation direction.
Effective modal mass	Mass contribution associated with a mode.
RHA	Response-history analysis. In this program, a linear modal educational postprocessing utility.
RSA	Response-spectrum analysis. In this program, a linear modal educational postprocessing utility.
ABSSUM/SRSS/CQC	Common modal response combination methods.
RCM	Reverse Cuthill-McKee style reordering concept used for bandwidth diagnostics where implemented.

Appendix B. Minimal source commands

The following commands summarize the source workflow. See the Installation Manual for complete installation instructions.

```
conda create -n CE4011 python=3.11 -y
conda activate CE4011
python -m pip install -r requirements.txt
python scripts\run_static_gui.py
python -m pytest -q
```

Appendix C. Executable distribution reminder

For the Windows distribution, keep the full dist\CE4011_Solver folder together. Do not copy only CE4011_Solver.exe. The _internal folder is required by the packaged runtime.

```
dist\CE4011_Solver\CE4011_Solver.exe
```